

QuizSmart - Program Flow

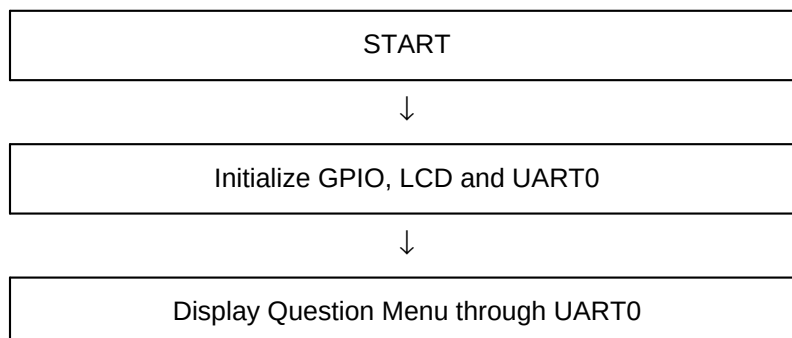
ARM7 LPC21xx Embedded Interactive Quiz System

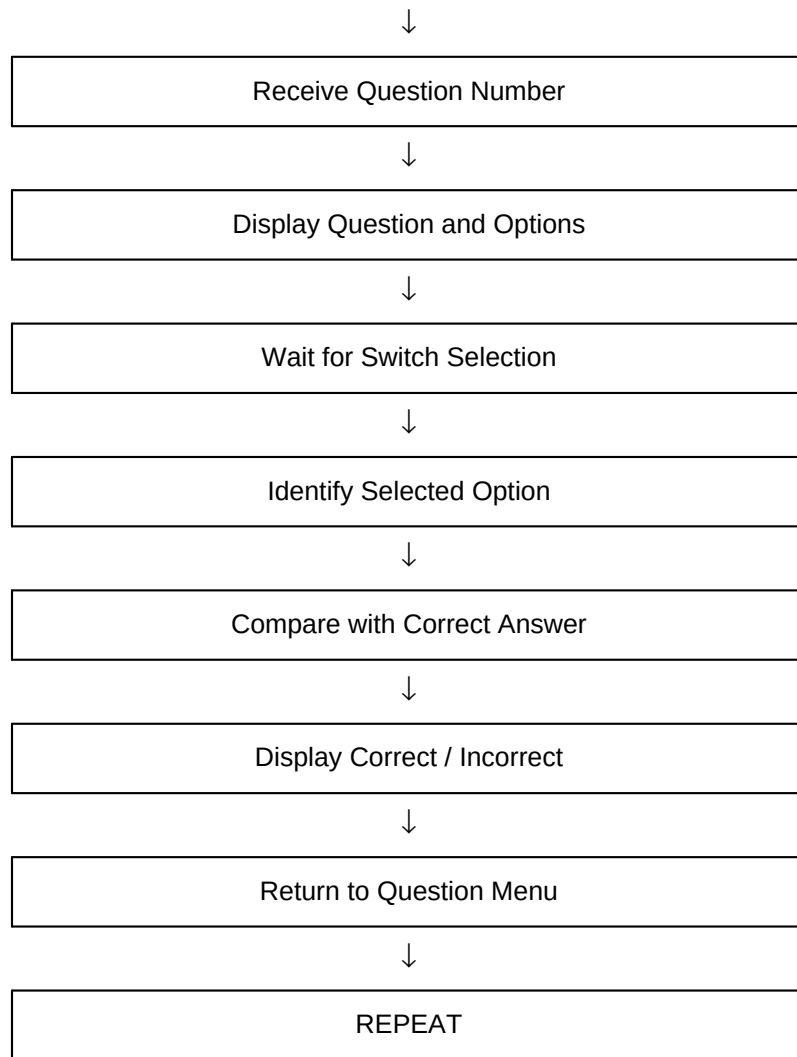
Program Flow Description

The program initializes the LPC21xx peripherals, presents a question menu through UART0, receives the selected question number, displays the question and options, waits for a push-button selection, verifies the answer, and returns to the main menu.

Step	Program Stage	Description
1	START	Program execution begins.
2	Initialize GPIO	P0.5 and P0.6 are configured as output pins for the LED interface. The switch inputs are read through GPIO.
3	Initialize LCD	The LCD driver is initialized to provide user instructions.
4	Initialize UART0	UART0 is initialized at 9600 baud for serial-terminal communication.
5	Display Question Menu	The available question numbers are transmitted through UART0.
6	Receive Question Number	The user enters a question number through the UART terminal.
7	Display Selected Question	The selected question and its four multiple-choice options are transmitted through UART0.
8	Prompt User	The LCD displays an instruction asking the user to select an option using the switches.
9	Poll Switches	The program continuously checks SW1, SW2, SW3, and SW4.
10	Identify Option	SW1, SW2, SW3, and SW4 correspond to options 1, 2, 3, and 4 respectively.
11	Verify Answer	The selected option is compared with the predefined correct answer for the selected question.
12	Display Result	UART0 displays either "correct" or "incorrect".
13	Return to Menu	The program returns to the question-selection menu and repeats continuously.

Simplified Flow





Implementation Note: The current main.c uses option1() in the question cases shown. The separate option2(), option3(), and option4() functions are defined but are not called by the displayed question cases.